

The Hand That Takes

*Hittite dā- as Pre-PIE *de + a, the Anatolian Directive Univerbation, and the Within-Hittite Chronology*

Paper 4 of the der- project. Builds on Chavis & Claude (2026a), (2026b), and (2026c).

[Working paper v5 — for scholarly review]

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Abstract

The first three papers of the der- project reconstructed pre-PIE *de- ‘hand’ and derived from it the body-part noun *de-r-, the contrastive suffix *-(t)ero-, the numeral *d̥uo-, and the give-verb *deh₃-. The present paper carries the framework into Anatolian, arguing that Hittite dā- ‘to take’ preserves a fossilized univerbation of pre-PIE *de- with a postposed directional element *a ‘toward the speaker,’ contracted by hiatus to long *ā in pre-Anatolian, at a stage before the verbal extension *-h₃- was added. The take/give polarity thereby resolves into a structural opposition: bare *de- in post-Anatolian PIE develops the give-verb via the *-h₃- extension; *de- extended by directional *a ‘toward the speaker’ gives the take-verb preserved in Hittite. The argument’s centerpiece is the Anatolian X+a parallel: *de + a > dā- fits a documented Anatolian morphological type rather than constituting an isolated proposal (anda < *en + a; katta < *k̥mt + a; parā < *pro + a; šarā < *uper + a). The Luwian lā- problem is the framework’s largest single liability and is engaged substantively in §9. Probability: 30–42%.

1. Introduction

Two facts about Hittite, taken together, demand engagement from the framework developed in Papers 1–3. The first is well known: Hittite dā- and cognate Luwian lā- reflect the same etymon as the give-verbs of the rest of the family but mean ‘to take,’ not ‘to give.’ The second is less often emphasized: Hittite preserves a productive synchronic system in which the bare da- stem combines with directional preverbs to specify deictic orientation — u- ‘hither’ yielding uda- ‘to bring here’ and pe- ‘thither’ yielding pēda- ‘to bring away’ (Hoffner & Melchert 2008: §20.7). The construction-type the framework hypothesizes — directional specification of the bare hand-verb by an attached element — is therefore not only reconstructable at pre-Anatolian depth but attested productively in Hittite on the same root.

The thesis of this paper is that bare *dā- ‘to take’ is itself the residue of a similar but older construction, in which a postposed directional element *a ‘toward the speaker’ attached to the bare hand-element *de- and underwent univerbation by hiatus contraction *e + a > *ā in pre-Anatolian. The verbal extension *-h₃- of Paper 3 is, on this view, a post-Anatolian derivation: added to bare *de- to create *deh₃- yielding the give-verbs of the other branches. Anatolian split off before that extension was added, retaining instead the older directive univerbation.

The central positive structural argument is the Anatolian X+a parallel (§3): *de + a > dā- fits a documented and productive Anatolian morphological type. The standard view treats dā- as an isolated irregular hi-verb with no productive class membership; the framework places it as a regular instance of the Set 1 directional univerbation type. The framework’s chronological architecture is independently supported by Norbruis (2021) on semantic grounds and by Bauhaus (2019) on morphological grounds. Against it stand two real costs: the within-Hittite chronology of -šten(i) (§8) and, more seriously, the Luwian lā- problem (§9), which the framework cannot fully resolve.

The paper proceeds as follows. §2 sets out the standard reconstruction and the synchronic u-/pe- system. §3 develops the X+a parallel as the central positive argument. §4 presents the specific proposal and its chronological architecture. §5 addresses Norbruis (2021). §6 covers Boley (2007) and the CAPERE iconographic evidence. §7 develops the PIE postpositional syntax context. §8 engages the within-Hittite chronology. §9 addresses the Luwian lā- problem. §§10–11 close with scope and the probability assessment.

2. The Standard Reconstruction and the Synchronic u-/pe- System

Kloekhorst’s (2008: 928–930) entry gives the canonical Old Hittite paradigm of dā-/d- ‘to take.’ The paradigm shows stem alternation between long-ā forms in the singular and short-vowel forms in the plural:

Form	Hittite	Notes
1sg pres.	<i>dāḥḥe / dāḥḥi</i>	strong stem dā-
2sg pres.	<i>dātti</i>	strong stem dā-
3sg pres.	<i>dāi</i>	strong stem dā-
1pl pres.	<i>tumēni, dāweni</i>	weak stem, /u/ or /aw/ vocalism
2pl pres.	<i>dattēni</i>	weak stem
3pl pres.	<i>danzi</i>	weak stem, /a/ vocalism

Table 1. The Old Hittite present paradigm of dā- ‘to take’ (after Kloekhorst 2008: 928–930).

Kloekhorst derives the paradigm from PIE *deh₃- with regular ablaut: the strong stem *dóh₃- gives long ā via the regular Anatolian outcome *ō < *oh₃; the zero grade *dh₃- gives /a/ in danzi via regular laryngeal vocalization (*dh₃-énti); the /u/ in tumēni comes from vocalization of zero-grade *dh₃-uēni. Melchert (1994: 89) reconstructs Proto-Anatolian *dō- for the take-verb and documents that long vowels in Anatolian can derive from contracted vowel-plus-laryngeal or vowel-plus-vowel sequences (§4.1.6.3.1). One observation from Kloekhorst is directly relevant to the framework: the compound verbs uda- ‘to bring here’ and pēda- ‘to bring away’ ‘preserve forms that are more archaic, e.g. petumen and utumen vs. dāwen, petišten vs. dāten’ (Kloekhorst 2008: 930). The paradigm of dā- is demonstrably innovative in its plural; the older state is accessible through the compounds.

Hoffner and Melchert (2008: §20.7) describe the synchronic Hittite u-/pe- preverb system: ‘In Hittite only two truly inseparable preverbs exist: u- “here” and pe- “there”, attested in the pairs: uda- “to carry here” and peda- “to carry away”; uiya- “to send here” and peya- “to send away”.’ The directional-reversal construction on the bare da- stem is not hypothetical: it is attested and productive. Under the standard reconstruction, the long ā of the simplex follows from o-grade laryngeal lengthening; under the framework, it falls out differently, as a contracted vowel from the fossilized postposed directional. The prefixed compounds on the same root preserve, in attested Hittite, the directional reversal construction that the framework hypothesizes for the simplex at pre-Anatolian depth.

3. The Anatolian X+a Parallel

The framework’s claim that Hittite dā- derives from a pre-Anatolian univerbation *de + a fits a documented Anatolian morphological type. This is the paper’s central positive structural argument: the proposal is not an isolated mechanism invented for this etymology but an instance of a productive Hittite class.

Hoffner and Melchert (2008: §20.3) describe the synchronic Hittite paradigmatic contrast between two paired sets of local adverbs. Set 1 (‘Place to Which’): anda ‘into,’ āppa ‘back,’ katta ‘down,’ parā ‘out, forth,’ šarā ‘upward.’ Set 1 ends in -a and marks direction-toward. Set 2 (‘Place Where’): andan ‘within,’ āppan ‘behind,’ kattan ‘below,’ pēran ‘in front,’ šer ‘above.’ Set 2 ends in -an and marks location-at. The formal alternation -a : -an maps consistently onto the directional/locative semantic contrast.

The Set 1 forms decompose transparently as a local element followed by the directional *-a: *en + a > anda; *kmt + a > katta; *pro + a > parā; *uper + a > šarā. The directional *-a is the

same morpheme as the Hittite allative case ending *-a* attested productively in nominal forms: *aruna* ‘to the sea’ (allative) vs. *aruni* ‘at the sea’ (locative); *nepisa* ‘to heaven’ vs. *nepisi* ‘in heaven.’ Across both the local-adverb system and the nominal case system, *-a* marks direction-toward consistently.

The framework’s **de + a > dā-* is structurally identical to **kmt + a > katta*. In each case, a local element is followed by the directional **-a* and contracts into a fixed form by hiatus. The contraction **kmt-a > katta* gives a geminated *tt* by regular vocalization of **m̥* before **t*; the contraction **de-a > dā* gives a long *ā* by hiatus contraction of **e + a*. Both contractions are documented in Anatolian phonology; both produce fossilized forms that have lost their original morpheme boundary. Melchert (1994: §4.1.6.3.1) documents that long vowels in Anatolian can derive from contracted vowel-plus-vowel sequences across morpheme boundaries, providing the general phonological principle the contraction requires. A fully specified sound rule for **e + a > *ā* specifically in verbal-stem position remains a phonological loose end, acknowledged in §10 below.

What is structurally distinctive about **de + a* — and what produced its surface as a verb rather than as an adverb or preverb — is that **de-* was, alongside its directional function, also a verbal stem in the polyfunctional pre-PIE morphology of Papers 1–3. The univerbation of a verbal **de-* with the directional **a* yielded a fossilized verbal stem rather than a fossilized adverb: the take-verb **dā-*, deictically reversed from the bare give-direction by the postposed ‘toward-oneself’ element. The standard view treats *dā-* as an isolated irregular *hi*-verb; the framework places it as a regular instance of a productive Anatolian class.

4. The Proposal and Its Chronological Architecture

At the earliest reconstructable stratum, pre-PIE was postpositional and OV (§7 below). At that stage, the bare hand-element **de-* functioned as a verbal stem, and a directional adverbial element **a* appeared postposed to it: **de- + *a → *de a → *dā-* (by hiatus contraction **ea > *ā*). The semantics compose: bare **de-* ‘extend (the hand)’ in the unmarked direction (= give, away from the speaker) plus directional **a* ‘toward the speaker’ yields ‘extend toward oneself’ = ‘take, receive.’

The construction is parallel to the synchronic Hittite *uda-* ‘to bring here’ (bare verbal stem + directional preverb ‘hither’), but with the directional element in postposed rather than prefixed position, reflecting the older morphosyntactic state. The framework makes a specific prediction about the plural forms of the paradigm: the forms with /u/-vocalism (*tumēni*) and /a/-vocalism (*danzi*) reflect the bare verbal stem **d-* without the directional **a*, attached to inflectional endings

whose vowels produced the surface forms. The long-ā of the singular then spread into the plural in late Old Hittite — exactly the innovation Kloekhorst documents, and equally well-predicted by the present analysis as by the standard one.

The four-paper framework’s chronological architecture identifies three derivational outcomes on *de- that Anatolian preserves, plus a fourth that is post-Anatolian:

Stratum	Derivational pathway	Surface form	Anatolian witness
Pre-Anatolian	bare *de- directional particle	*de/do	Hittite Set 1 directional -a; allative case -a
Pre-Anatolian	*de- + *-ų- + *-o/-i-	*du-o-/du-i-	Hittite dāyu- ‘two’ (standard)
Pre-Anatolian	*de- + *a (directive univertation)	take-verb *dā-	Hittite dā- ‘to take’ (this paper)
Post-Anatolian	*de- + *-h ₃ - (verbal extension)	give-verb *deh ₃ -	absent from Anatolian

*Table 2. The chronological architecture of the four-paper framework. A horizontal rule separates the three pre-Anatolian derivational outcomes from the one post-Anatolian outcome. The post-Anatolian give-verb *deh₃- is absent from Anatolian because the *-h₃- extension was added after the Anatolian split.*

Bauhaus’s (2019: §5) independently published paradigmatic opposition between locative *-r and directive *-o on the same nominal roots provides published external precedent for the structural type the framework reconstructs on *de-: the locative member *de-r- was fossilized as a body-part noun before the Anatolian split; the directive member *de + a remained productive into Anatolian, where it contracted to *dā- and was preserved as an active verb. Bauhaus’s chronological observation that the directive ‘could keep its productivity at least until Anatolian’ while the locative *-r ‘ceased to be productive in all attested languages’ is precisely the asymmetry the framework predicts.

5. Norbruis (2021): Independent Semantic Support

Norbruis (2021: 243–249), in his Leiden dissertation on Anatolian morphology and semantics, argues independently for a chronology closely aligned with the morphosyntactic chronology proposed here. His conclusion: ‘In this chapter it is argued that the verb *deh₃- did not mean “to give” in PIE, but “to take,” as in Anatolian. Although the verb did not survive in any evident way in Tocharian and Germanic, the semantic shift to “to give” is likely to be an innovation of post-Anatolian IE.’

Norbruis works through every parallel adduced for a give > take semantic development and argues they actually support the opposite direction. He reconstructs a semantic chain: ‘to take’ > (with a goal or beneficiary constituent) ‘to bring, convey, deliver’ > (when the recipient is animate) ‘to hand over, to give.’ The intermediate stage — physical delivery of concrete objects to animate recipients — is directly attested: Eska’s (2004) analysis of the Prestino inscription (northern Italy, 5th–6th c. BCE) identifies the formula as describing the physical delivery of concrete objects (seats) to animate divine recipients — exactly Norbruis’s reconstructed intermediate stage, not yet the fully abstract ‘give’ of later Celtic usage.

Norbruis’s argument is fully independent of the present project and does not require any of its formal proposals. His semantic chronology and the framework’s morphosyntactic chronology are decoupleable: each can stand or fall on its own evidence. The coincidence of formal and semantic chronology is mutually reinforcing rather than circular; the two analyses converge on the same pre/post-Anatolian divide from different directions.

6. Boley (2007) and the CAPERE Iconographic Evidence

The proposal that *deh₃- derives from a directional particle has a historiographic precursor. Boley (2007) previously proposed deriving the give/take root from ‘a directionally ambivalent particle reconstructed on the basis of Latin *dē* “from” and Greek *-δε* “to”’ (as reported by Norbruis 2021: 245 n.9). The same Greek allative clitic *-δε* that Paper 3 adduces for *de- appears in Boley’s earlier proposal. Norbruis dismisses Boley’s specific arguments as unsound, and we agree with much of that criticism: a ‘directionally ambivalent’ particle with both ‘from’ and ‘to’ senses simultaneously is not well motivated. What survives is the structural insight that the give/take verb derives from a directional particle. The present paper offers a more constrained version: a single directional-toward element *a combined with the bare hand-element *de- to yield the take-meaning, while the bare element alone in post-Anatolian developed the give-meaning through the *-h₃- extension.

Anatolian hieroglyphic sign L 41 (Laroche numeration) is iconographically a ‘taking’ or ‘grasping’ hand. It functions both as the logogram CAPERE ‘to take’ — used to write words for taking, most commonly the Luwian verb *lā-* — and as a syllabogram with phonetic value /tā/ (Rieken 2008; Yakubovich 2008). Yakubovich (2008) derives the syllabic value /tā/ acrophonically from Hittite *dā-* ‘to take’ in the bilingual Hittite-Luwian environment of script development. Under the present framework — where *dā-* derives from pre-PIE *de- (the hand-element) + postposed directional *a — the hand iconography of L 41 preserves the verb’s etymological source. Kloekhorst (2019) offers a competing analysis, deriving the sign’s value from Luwian *lā-* reconstructed as PA *dh₃- with a preserved laryngeal; this re-analysis depends on

the contested phonological framework engaged in §9. Yakubovich’s acrophonic analysis does not depend on contested mechanisms; we treat it as the more parsimonious basis for interpreting the iconographic evidence. The iconographic argument is corroborative, not load-bearing.

7. PIE Postpositional Syntax

The framework’s proposal requires that pre-PIE allowed postposed adverbial directional elements to appear after verbal stems. This is the unmarked configuration for the earliest reconstructable layers of IE syntax. Lehmann’s (1974) *Proto-Indo-European Syntax* argued on typological grounds that PIE was an OV language whose unmarked phrasal pattern required postposed adpositions and modifiers; Friedrich (1975) refined the typology. The Anatolian and comparative evidence supports this picture.

Hittite is uniformly postpositional in its adpositional system: all Hittite adpositions are postpositions in nominal phrases (parni anda ‘into the house,’ with anda following the dative-locative parni). As the earliest-attested Indo-European branch, Anatolian’s postpositional pattern is a strong indicator of the PIE state. Tocharian, the second-earliest branch to split off under most current chronologies, is also predominantly postpositional. Vedic Sanskrit preserves substantial postpositional flexibility. Latin preserves postpositional archaisms in fossilized form: mēcum ‘with me,’ tēcum ‘with you,’ nŕbiscum and vŕbiscum all have cum postposed to the personal pronoun.

Homeric Greek shows the relevant change in progress. In Homer the so-called ‘preverbs’ remain separable from the verb (tmesis: katà dákrua leíbwn ‘shedding tears down,’ with katá free-standing rather than fused into Classical katalEíbwn). By Classical Greek these elements have univerbated with the verb stem. The construction the framework proposes for the bare hand-element — *de- + postposed directional *a > fossilized *dā- — is a special case of this documented PIE pathway, occurring at an earlier stratum.

8. The Within-Hittite Chronology

Kloekhorst (2007) documents that the replacement of the original ħi-conjugation 2pl ending -šten(i) by the mi-conjugation -tten(i) in Hittite happens in three chronological waves: Wave 1 (pre-Hittite) affects consonant-final stems; Wave 2 (pre-Hittite to Old Script) affects laryngeal-final stems; Wave 3 (Middle Hittite to New Hittite) affects vowel-final stems. The simplex dā- already shows -tten in OS texts (da-at-te-e-ni, da-a-at-te-en) — it patterns with laryngeal-final stems that received the replacement in Wave 2, not with vowel-final stems that received the replacement only in Wave 3.

Under the framework’s analysis, where *dā-* derives from **de + a* by hiatus contraction and is therefore a vowel-final stem, the simplex should pattern with Wave 3, not Wave 2. The standard view, which derives *dā-* from **deh₃-* with a root-final laryngeal, predicts Wave 2 patterning directly. This is a real cost for the framework.

Two mitigating considerations are available. The first is the OS hapax *pe-tí-iš-te-en* in the compound verb *peda-* ‘to bring away.’ This form preserves two archaic features the simplex has lost: the *-šten* ending itself (the Wave 1/2 archaism) and a weak stem *ti-* not attested in the simplex paradigm. The compound therefore preserves an older paradigm shape than the simplex, exactly as the framework predicts for a fossilized univerbation: the simplex has been leveled and lexicalized as a unitary stem, while the compounds preserve traces of an older state. Kloekhorst’s own observation — that the compounds show forms ‘more archaic’ than the simplex — is the published support for this reading.

The second consideration is analogical: the framework’s hiatus contraction **e + a > *ā* produces a phonologically heavy syllable that may have been classified morphologically with stems ending in a long vowel from laryngeal coloring (**-eh₂ > *-aH > *-ā*). By the time of attested Hittite, the original morpheme boundary of **de + a* was no longer detectable; the verb behaved as a unitary stem with a long *ā* and may have been attracted into the laryngeal-final morphological class by surface analogy. The standard view fits the *-šten(i)* datum more cleanly; the framework’s accommodation through analogical reclassification is available but less parsimonious.

9. The Luwian *lā-* Problem

The Luwian *lā-* problem is the framework’s largest single liability. The standard view (Kloekhorst 2008, 2013, 2019; Norbruis 2021) derives Luwian *lā-* ‘to take’ from PA **doH-* / **dH-* with **h₃* preserved as glottalization of the initial stop, and subsequently developing to a lateral in Luwian via the sound change **d^ʰ- > Luwian l-*. Kloekhorst (2019) makes this argument explicitly and uses it to explain the value of Anatolian hieroglyph L 41 (§6 above). The framework does not propose a positive Luwian-internal etymology for *lā-*; this is a genuine gap.

Three considerations limit the weight that should be placed on the standard derivation. First, the proposed sound change **d^ʰ- > Luwian /l-/* has no independent parallels elsewhere in Luwian or Anatolian phonology. The documented Luwian sound change involving *l* is rhotacism: in Iron Age Hieroglyphic Luwian, intervocalic *d, l, n > r* (e.g. **īdi > īri* ‘he gets’; *wala- > wara-* ‘die’; Yakubovich and Shatskov). This change goes the wrong direction for Kloekhorst’s derivation and operates intervocalically, not word-initially. There is no general sound law producing *l-* from word-initial **d-* in Luwian; the derivation is accordingly ad hoc, invented for this single

etymology.

Second, the glottalization framework on which Kloekhorst's derivation depends is contested. Melchert (2015) argues explicitly: 'There is no evidence for word-initial $*h_1$ preserved as [ʔ] in Hittite and Luvian, contra Kloekhorst (2004, 2006a, 2008) and Simon (2010, 2013).' Sideltsev and Molina (2021) raise independent methodological objections to Kloekhorst's broader prosodic-phonological framework. Third, $*h_3$ has the weakest independent evidence base of the three PIE laryngeals: $*h_1$ leaves demonstrable e-coloring; $*h_2$ leaves clear a-coloring and compensatory-lengthening patterns; $*h_3$ reduces primarily to o-coloring in Greek and the contested Anatolian glottalization argument. Kloekhorst's derivation of Luwian *lā-* stands at the most vulnerable point of laryngeal theory.

The framework's position is as follows. The standard derivation has no general sound law to support it, depends on a contested phonological framework, and rests on the most vulnerable point of laryngeal theory. These are reasons not to treat the standard derivation as a decisive argument against the framework's pre-Anatolian chronology. But the framework does not have a positive Luwian-internal etymology to offer as an alternative. The standard analysis has the advantage of formal economy within the laryngeal-theoretic framework and derives Hittite *dā-* / Luwian *lā-* from a single PIE root with established sound laws, even if those laws are contested. The question of where Luwian *lā-* comes from remains genuinely open, and the framework registers this openly as its principal unresolved difficulty.

10. Scope and Remaining Problems

Three problems deserve explicit statement. None is fatal, but two are real.

The Luwian *lā-* problem (§9) is the dominant liability. The framework does not propose a positive Luwian-internal etymology, and the absence of one is a genuine gap. The mitigating consideration — that the standard derivation is itself phonologically ad hoc and rests on a contested framework — weakens the standard derivation as an argument against the framework, but does not supply what the framework is missing.

The within-Hittite chronology (§8) is a softer but real cost. The simplex *dā-* patterns with laryngeal-final stems rather than vowel-final stems. The compound *peda-* preserving the OS hapax *petišten* with its archaic weak stem *ti-* is a positive datum: the compound preserves features the simplex has lost, exactly as the framework predicts for a fossilized univerbation. The simplex's Wave 2 patterning is accommodatable through analogical reclassification, but the standard view fits the datum more cleanly.

The hiatus contraction $*e + a > *\bar{a}$ (§3) requires sharper phonological specification. The Anatolian X+a paradigm provides cross-form support (parā and šarā show long ā from analogous structures; the nominal allative -a is the same morpheme), and Melchert (1994) documents the general principle that long vowels can arise from vowel-plus-vowel contractions across morpheme boundaries. A fully specified sound rule for $*e + a$ in verbal-stem position remains a phonological loose end.

Whether other Hittite ḫi-verbs with isolated long ā might be analyzable as fossilized univerbations of bare verbal stems with postposed directional elements is a question the framework raises but does not address here. If $*de + a$ is not the only example of the construction, additional cases would strengthen the proposal; their absence would not refute it, since the fossilized univerbation is predicted to be rare.

11. Probability Assessment

Conditional on Papers 1, 2, and 3 being correct, we assess the probability of the central thesis of this paper — that Hittite dā- ‘to take’ preserves a pre-Anatolian directive univerbation $*de + a$ — at approximately 30–42%. This is a downward revision from earlier estimates, reflecting honest accounting of the Luwian lā- problem and the within-Hittite chronology cost.

In favour: the X+a parallel is the framework’s central constructive argument, and it is genuinely structural rather than merely analogical — $*de + a > dā-$ is one instance of a documented Anatolian morphological type, not a one-off proposal. The chronological architecture of Table 2 is internally consistent and aligns with two independently published positions: Norbruis (2021) on the semantic chronology and Bauhaus (2019) on the locative/directive paradigmatic pairs. The synchronic u-/pe- prefix system confirms that directional reversal of the take-verb by an attached element is real and productive on the same root in attested Hittite. The OS hapax petišten in peda- is exactly the fossilized-univerbation residue the framework predicts. Against: the Luwian lā- problem has no positive resolution; the within-Hittite chronology cost is real; and the hiatus contraction needs sharper specification.

The marginal probability for the four-paper framework as a whole — integrating the conditional probabilities: 30–40% (Paper 1) $\times$ 35–45% (Paper 2) $\times$ 35–45% (Paper 3) $\times$ 30–42% (Paper 4) — is approximately 1–4%. Much of the evidence supports multiple strata simultaneously rather than each stratum independently, so the strata are not fully independent; the honest marginal range is toward the lower end of that interval. We regard the central claim as more likely wrong than right at the marginal level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

Authorship and Contributions

Chavis's contributions. The hypothesis that Hittite *dā-* 'to take' preserves an earlier construction in which the bare hand-element **de-* was extended by a postposed directional element originates with Chavis. All decisions about scope, framing, and inclusion are Chavis's. The decision to foreground the X+a parallel as the load-bearing positive argument — making the Anatolian morphological type the constructive center rather than background context — is Chavis's. The decision to engage the Luwian *lā-* problem substantively rather than rely on the iconographic argument alone is Chavis's. The decision to address the within-Hittite chronology is Chavis's, made after identifying the *-šten(i)* data as genuinely informative about the framework's prediction structure. The hypothesis that PIE postpositional syntax dissolves the central morphosyntactic objection to the framework's construction is Chavis's. All fundamental decisions about scope, framing, and direction across all four papers rest with Chavis, who supplies all intellectual continuity through systematic transcript management.

Claude's contributions. Claude reviewed prior-thread material on the Luwian *lā-* problem (Kloekhorst 2019; Melchert 2015; Sideltsev & Molina 2021) and on the within-Hittite chronology (Kloekhorst 2007 on *-šten(i)*); searched for documented Luwian sound changes involving *l* (finding rhotacism as the only general change, operating in the wrong direction for Kloekhorst's derivation); drafted the §3 exposition of the X+a parallel as the central positive structural argument; drafted the §8 engagement of the within-Hittite chronology with the *peda-/petišten* compound as mitigating positive evidence; drafted the §9 substantive engagement of the Luwian *lā-* problem; and drafted the §4 architectural convergence. Claude additionally read Kloekhorst (2008, 2013, 2019), Hoffner & Melchert (2008), Norbruis (2021), Bauhaus (2019), Boley (2007), Yakubovich (2008), Rieken (2008), Melchert (1994, 2015, 2017), Lehmann (1974), and Friedrich (1975). Claude drafted all prose sections from Chavis's structural and substantive direction. The probability estimate in §11 is Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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